ACOUSTIC NODE CARRIER — INSPECTION PACKET

2-layer CNC isolation-milled · 56 x 53 mm · unplated holes · GND pour (bottom)

LAYER / ROUTING

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RED  = top copper   (SPI bus, mic I2S, NRST, C1 decoupling AT LoRa VDD)
BLUE = bottom copper (3V3 rails + JP1 ferrite + threaded crossing, PPPS) + POUR
GRN  = 0 jumper wires!  BUSY = top->via->bottom copper; DIO1 = all bottom copper
0 copper crossings + 0 trace-over-pad shorts (machine-verified DRC).
3V3: LDO feeds mic/GPS with no crossing; JP1 ferrite, then ONE 0.6mm trace
     threads the ESP RX<->G1 gap (pads 1.5mm) to feed ESP-3V3, LoRa VDD, C2.
EDGE CUTS: cut the outline + USB-C plug notch (bottom) + coil-ant notch (top)
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BILL OF MATERIALS (per node; blue rows are clickable -> Amazon)

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  U1   ESP32-S3 SuperMini - soldered FLAT via castellations (not socketed)
  U2   INMP441 I2S mic - O13.9 round, 2x3 @ 2.54, rows 7mm apart
  U3   ATGM336H GPS (1x5) - hangs off TOP edge  [confirm PPS pad on listing]
  U4   REYAX RYLR689 LoRa (LLCC68 SPI, 915MHz, 1.27mm castellated), 2pk
  U5   MCP1700-3302E/TO LDO 3V3 TO-92 (1 GND|2 VIN|3 VOUT)  2nd:HT7333-A
  C1   10uF 25V MLCC 0805 at LoRa VDD (top-mount, ties to LVDD/LGND)
  Cin  1uF + 10uF 0805 MLCC assortment kit (Cin/Cout, 1uF each)
  C2   470uF 10V/16V LOW-ESR electrolytic (Nichicon PM) = TX-spike buffer
  L1   ferrite bead 0805 ~600ohm@100MHz (TDK MPZ2012S601) on JP1 3V3 crossing
  V1   1 via (BUSY). JP1=0805 via-in-pad: solder-bridge to rail; bridge pads=bypass
  BAT  18650 holder (2-wire) + your CN3791 charger -> BATT-IN pads (3.0-4.2V)
  HW   4x M3 standoff/screw kit   +   2-layer FR1/FR4 clad stock
  2nd-source LDO: HT7333-A 20pk (TO-92, same pinout)
  L1 isolates ESP+LoRa HF noise from the mic/GPS supply (~20-40dB, 50-300MHz). It
  does NOT fix TX droop (C2 does) or radiated GPS<->LoRa antenna coupling. Optional:
  a solder bridge across the pads bypasses it (0-ohm) if it ever causes trouble.
  NOTE: coin SUPERCAPS rejected -- 30-120ohm ESR drops 4-9V on the 120mA TX pulse;
  a 470uF low-ESR e-cap (<0.1ohm) sources it flat.
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NET MAP (matches firmware node_config.h)

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  GP4  Mic BCLK   GP5  Mic WS    GP6  Mic SD     (I2S, L/R->GND = left)
  GP13 MISO  GP12 MOSI  GP11 SCK  GP10 NSS       (SPI, ESP right column)
  GP8  RFSW_V2    GP9  RFSW_V1   (RF switch; each pin to its outer pad)
  GP7  BUSY (copper+via)   GP3 DIO1 (copper)   GP44 NRST (top)
  GP1  GPS TX(in) GP2  GPS PPS   (GPS RX = no-connect)
  3V3 -> mic/gps/lora VDD (C1 at LoRa VDD)  GND pour -> all grounds
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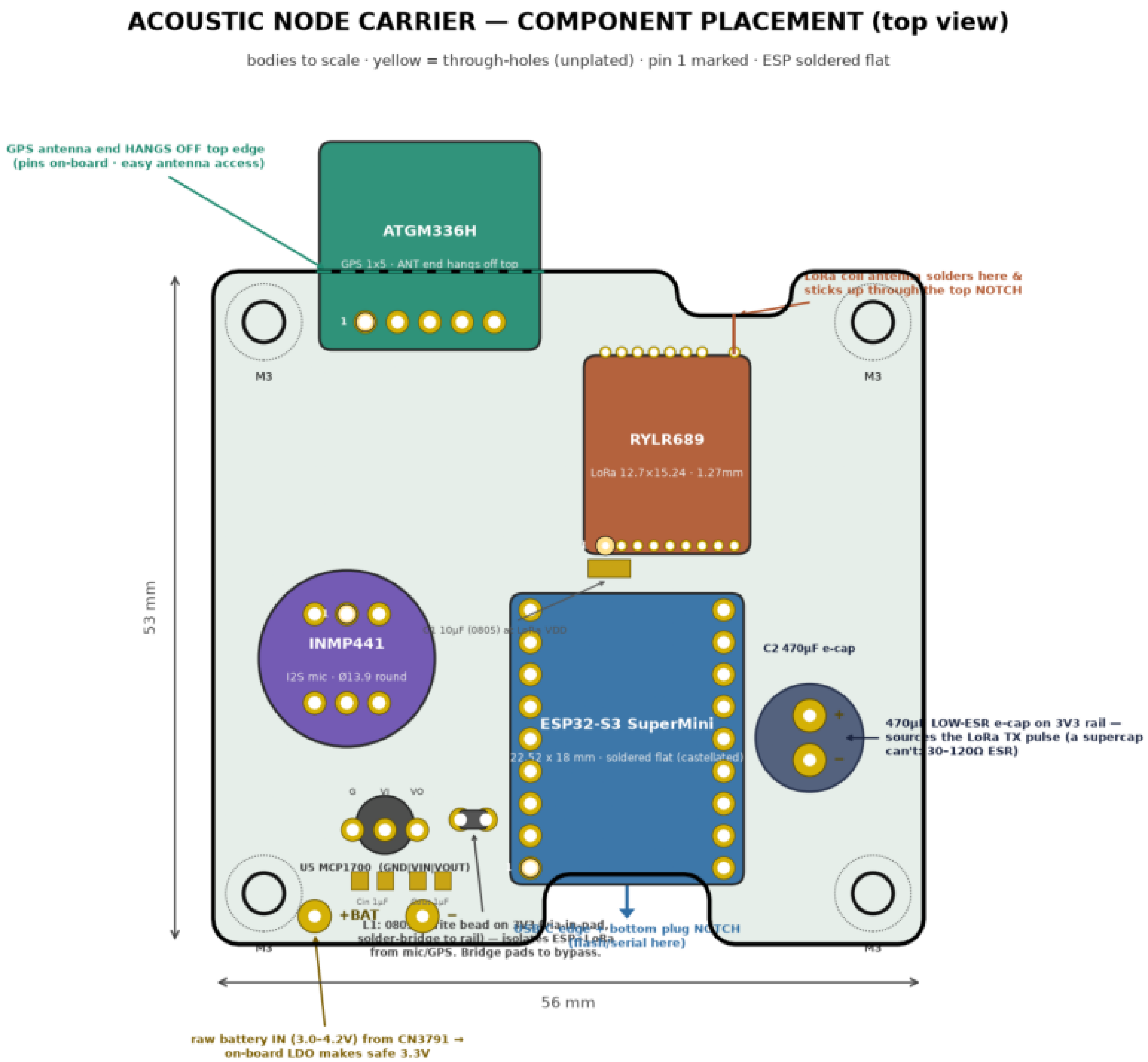
>> HOW TO CHECK PARTS FIT (before cutting copper) <<

1. Print PAGE 4 at 100% / "actual size" (NOT fit-to-page). Then caliper the 50 mm scale bar — it MUST read 50.0 mm. Re-print if off.
2. Set each real module on its pad group: pin pitch lands in the holes, the body outline (dashed) has clearance, antenna/USB edges point the right way.
3. Caliper-verify the footprints assumed vs your actual parts:
    - ESP SuperMini 2.54 pitch, rows 15.24, body 22.52x18
    - RYLR689 body 12.70x15.24, pads 1.27 pitch, rows 15.24 <- finest
    - ATGM336H 1x5 @ 2.54   ·   INMP441 Ø13.9 round, 2x3 @ 2.54, rows 7mm
4. DRILL-ONLY test cut on scrap; dry-fit ALL parts + standoffs; THEN mill.
5. Optional: mill one board first, populate + verify before making the 2nd.

pages: 2 = placement (assembly)   3 = routed copper   4 = 1:1 fit/drill template

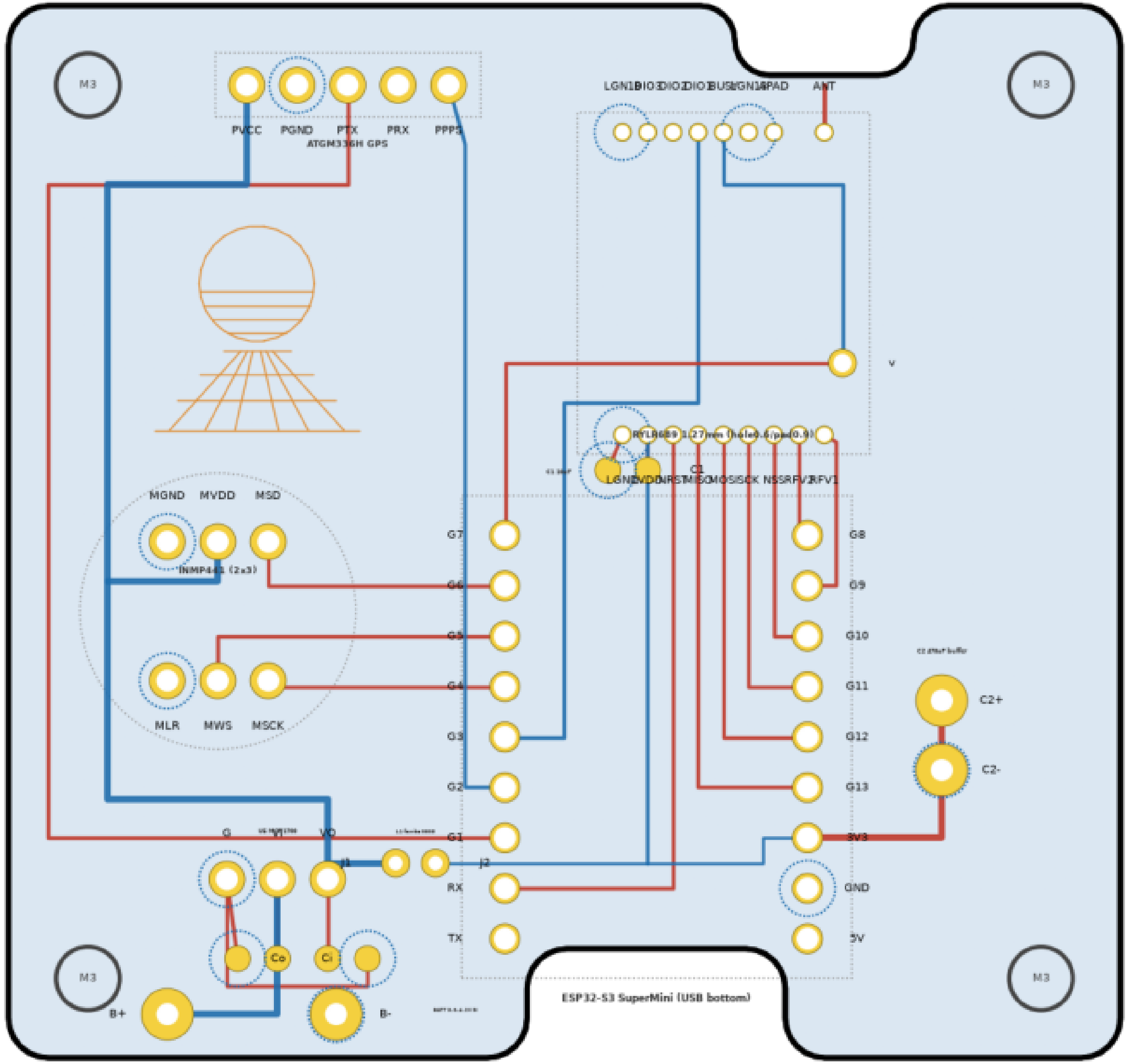
PAGE 2 — COMPONENT PLACEMENT (assembly, top view)



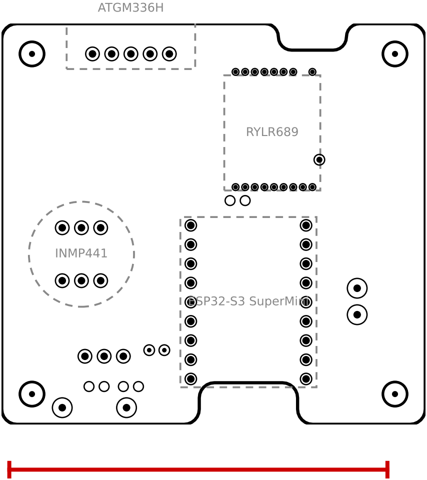
PAGE 3 — ROUTED COPPER (red=top, blue=bottom, green=jumper)

ACOUSTIC NODE CARRIER · 2-layer milled · 56x53 mm

RED=top copper   BLUE=bottom copper   GREEN dashed=jumper wire   light-blue field=GND pour   ·   0 copper crossings



4 — 1:1 FIT / DRILL TEMPLATE (solid dots = drill points, dashed = module b



50.0 mm — PRINT AT 100%; this MUST measure 50 mm with calipers